Modelling Moon Phases · Teacher and Cover Plan

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Scheme Outcome

TRACK Science scheme of work: Earth and Space

Use a physical 3D model to investigate and record the sequence of 8 Moon phases from Earth's perspective and evaluate the model.

Success Criteria

I can map each part of the torch, observer, and ball model to the Sun, Earth, and Moon.

I can record the 8 Moon phases in order from Position 1 (New Moon) to Position 8.

I can explain one strength and one limitation of our classroom Moon-phase model.

Model Sequence (I do)

First, stand on the 8-point floor mat with Position 1 pointing straight toward the fixed lamp/torch.

Second, turn anticlockwise from Position 1 to Position 8, holding the ball at arm's length and observing only the side facing you.

Third, shade the dark fraction and leave the lit fraction white in each of the 8 phase circles in your book.

Made by Matt

Our 3D ball-and-torch model has a big strength: it proves in 3D that half the ball is always lit while the observer's view changes.

However, our model has limitations: the torch beam spreads out and is much smaller than the ball, whereas the real Sun is gigantic and far away.

Also, the observer's head stays still or turns in steps, whereas the real Earth spins once every 24 hours while orbiting the Sun.

We Do & Hinge Check

An observer records a First Quarter Moon on Day 7. Which two phases will they observe on Day 10 and Day 14?

Answer: A. Between First Quarter (Day 7) and Full Moon (Day 14), the lit share continues growing through Waxing Gibbous.

Safe Practical & Card Setup

Use the 8-position Moon-phase model (or the 8-position observation cards) to shade and label the complete lunar cycle.

Kit: Fixed directional lamp/torch (at desk height), white ball on a short dowel, and 8-position floor/desk mat.

Keep the torch pointed horizontally at the ball, never into anyone's eyes. Keep the floor around the mat clear.

You Do Routes (Supported, Independent, Aspire)

Supported: Match the 8 phase pictures to Positions 1 to 8 on your sheet and label New Moon, First Quarter, Full Moon, and Third Quarter. (Help: Pos 1 = all dark (New Moon). Pos 3 = right half lit (First Quarter). Pos 5 = all lit (Full Moon). Pos 7 = left half lit (Third Quarter).)

Independent: Complete the 8-position Moon phase diagram (shading the dark part at each position), label the phases, and state one strength and one limitation of the ball-and-torch model. (Help: From Pos 1 to Pos 5 the lit part grows (__); from Pos 5 to Pos 1 it shrinks (__). A strength of the model is __, while a limitation is __.)

Aspire: Calculate the approximate number of days between each of the 8 model positions in a 28-day cycle, and evaluate two geometric limitations of the ball-and-torch model. (Help: 28 days divided by 8 positions is __ days per step. Two limitations are that the torch is smaller than the __ and the distances are not to __.)